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Assignment: Gene Expression Analysis and Interpretation
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Assignment 2 - Gene Expression Analysis and Interpretation

Introduction:

The HER2 receptor (termed ERBB2 for the remainder of the paper) plays an important role in normal cell growth and differentiation across different tissues within the human body. ERBB2 is part of the ERBB receptor family where the most notable other member of the family is EGFR. Normal functionality is characterized by a small number of ERBB2 molecules that inevitably results in the formation of a smaller number of heterodimers that are formed.¹ This small number of heterodimers is what leads to growth signals that are weak and therefore controllable.² Due to ERBB2's signaling and its most common pairing through dimerization with HER3, the subsequent pathways that can be affected include the MAPK and PI3K-AKT, respectively. In contrast to normal functionality of ERBB2, when ERBB2 is mutated and overexpressed, there is a greater number of heterodimers formed than normal and the resulting cell signaling is stronger ultimately resulting in malignant growth.³ This amplification and malignant growth in the breast area is the cause of one of the more aggressive forms of breast cancer. According to *HER2-positive breast cancer* by Loibl et Al., HER2 positive breast cancer accounts for around 15-20% of breast cancers.⁴ The current options for therapy treatment with ERBB2 amplified breast cancer have been well acknowledged for metastatic and early HER2-positive breast cancer, most notably with some success using the monoclonal antibodies drugs pertuzumab and trastuzumab in addition to chemotherapy.⁵ However, this success is limited as pointed out in *HER2-positive breast cancer* due the high death rate that still exists at the time this

¹ Rubin, I. and Yarden, Y. (2001) "The basic biology of HER2," *Annals of Oncology*, 12, pp. S3–S8. Available at: https://doi.org/10.1093/annonc/12.suppl_1.S3.

² (Rubin and Yarden, 2001)

³ (Rubin and Yarden, 2001)

⁴ Loibl, S. and Gianni, L. (2017) "HER2-positive breast cancer," *The Lancet*, 389(10087), pp. 2415–2429. Available at: [https://doi.org/10.1016/S0140-6736\(16\)32417-5](https://doi.org/10.1016/S0140-6736(16)32417-5).

⁵ (Loibl and Gianni, 2017)

paper was published with ERBB2 amplified breast cancer.⁶ Therefore the goal of this analysis and interpretation is to further investigate the effects of ERBB2 in breast cancer through the lens of amplified expression levels and non-amplified expression levels to understand more about the biological processes and pathways that are subsequently affected.

Methods:

Before performing any analysis, the Breast Invasive Carcinoma TCGA PanCancer data set was downloaded from cBioPortal. This dataset included a wealth of information that is specific to Breast Invasive Carcinoma, however there were three specific files that were used: data_mrna_seq_v2_rsem.txt (RNA-sequencing data), data_clinical_patient.txt (clinical patient data), and data_cna.txt (copy number alteration data). All features from the RNA sequencing and copy number alteration data files were used which included the gene symbol and Entrez ID, in addition to the expression levels for each of the patient samples / patients. However, in the data_clinical_patient.txt file, the following features were extracted and used from the file:

- patient ID
- OS Status (Overall survival)
- OS Months (Overall survival in months)

Before any analysis was done, the data was grouped by two characteristics being ERBB2 amplified or ERBB2 non-amplified based on their CNA level being greater than zero or less than zero, respectively, across the patients that existed from all three datasets. The raw data of the RNA-sequencing file was then cleaned to account for any missing values, including any genes that had larger than three features missing. The combination of dplyr (Wickham *et al.*, 2023) and basic R, in addition to tidyr (Wickham, Vaughan and Girlich, 2024) were used during the cleaning and preparation process. After the data was organized by amplified and non-amplified groups, the differential expression using the normalized and cleaned RNA-seq data across all matching patients was performed using the DESeq2 package (Love, Huber and Anders, 2014). This allowed for further analysis through identification of biological functions and pathway enrichment descriptions with the most significantly differentially expressed genes used through the enrichGO

⁶ (Loibl and Gianni, 2017)

(Yu, 2025) and the Gene Ontology databases (Carlson, 2025) as well as the KEGG and Reactome databases (Yu and He, 2016). Extraction of the variance stabilized transformed (vst) values using DESeq2 was performed and then displayed through the use of PCA plotting and heatmaps (Kolde, 2025). Finally, the vst values were used in a survival analysis through the generation of an overall survival model using glmnet's coxnet model regression (Simon *et al.*, 2011).

Interpretation of Results from the Analysis:

Differential Expression with HER2 Amplified / Non-Amplified – Header 1.1

ERBB2_Status	Count
Non_Amplified	740
Amplified	328

```
Key:
LFC -> Log Fold Change
p.adj -> p adjusted value

Results:
Total Over-Expressed Significant Genes(LFC > 0 ): 4186
Total Under-Expressed Significant Genes (LFC < 0 ): 5583
Total Significant Genes (p.adj < 0.05): 9769
```

The figures above represent the results from the overall differential expression analysis. These figures display results from the differential expression of the ERBB2 gene in reference to the total count of all the amplified genes in addition to the numbers of over-expressed, under-expressed, and total significant genes that were identified amongst the matched patients in the datasets. The two values being used for comparison in the over/under expression and total number of significant genes, the p-adjusted value (< 0.05) and log fold change (LFC $>$ or < 0) value, were chosen to subset the genes that are statistically significant and that experienced a change in expression levels greater than normal.

The status of ERBB2 being amplified or not was determined based on the copy number aberrations level that was recorded in the data_cna.txt file, where a CNA level greater than zero was categorized as amplified as opposed to a CNA level less than or equal to zero which was

categorized as non-amplified. The result was a significant proportion of patients seen to have an ERBB2 amplification (30.71%). This means that a significant number of patients were found to have an increased expression of ERBB2 through their tumor samples. The importance of understanding the number of patients who experienced an amplification in their ERBB2 levels is that this allows for the percentage of patients that have the ERBB2 amplification and the percentage of patients that do not have the amplification to be known. This will be a baseline understanding for downstream interpretations such as the pathway enrichment analysis that follows.

The over-expressed and under-expressed genes were extracted by taking the differentially expressed genes from ERBB2 that were significant according to their respective p-adjusted values and log fold change values in terms of the thresholds described above. The proportion of over-expressed significant genes was 42.85% compared the under expressed significant genes which were 57.15% of all significantly differentially expressed genes. The total count of the significant differentially expressed genes identified was 9,769. These results are important because the statistically significant over-expression or under-expression of genes can play significant roles in these patients with breast cancer when looking at each genes individual function.

Top 10 differentially expressed Genes ranked by fold change – Header 1.2

```
Key:
Gene -> identifier for the gene
baseMean -> average expression level for gene across all samples
log2FoldChange -> effect size and direction of the expression level change
lfcSE -> standard error of log2FoldChange
stat -> Wald test statistic score
pvalue -> statistically significance
padj -> adjusted p value to consider many statistical tests applied

Results:
ERBB2_Status_Amplified_vs_Non_Amplified
log2 fold change (MLE): ERBB2 Status Amplified vs Non-Amplified
Wald test p-value: ERBB2 Status Amplified vs Non-Amplified
```

Gene	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
CSN2	18.52793	-4.56354	0.845434	-5.39787	6.74381e-08	5.32552e-07
SPANXA2	2.34148	4.35182	0.636206	6.84028	7.90405e-12	1.33794e-10
GAGE12D	10.26094	4.29677	0.796620	5.39375	6.90016e-08	5.43224e-07
SPANXC	2.42993	4.17245	0.524362	7.95721	1.75964e-15	5.73287e-14
GAGE2B	1.52261	4.05698	1.412768	2.87165	4.08332e-03	9.81801e-03
CSN3	47.14761	-3.70971	0.492304	-7.53540	4.86839e-14	1.25265e-12
FAM9C	1.69449	3.37450	0.297923	11.32676	9.67139e-30	1.32292e-27
PNMT	165.00060	3.30959	0.179624	18.42513	8.25934e-76	5.90956e-73
GAGE4	4.45378	3.30569	0.670901	4.92725	8.33954e-07	5.10330e-06
LALBA	22.78384	-3.29068	0.729214	-4.51265	6.40239e-06	3.18629e-05

The top ten genes seen above are from the differential expression analysis of ERBB2 amplified or non-amplified, where the explanation of how the differential expression was done, what was being compared, and the overall set up for the analysis is discussed in the explanation provided under Header 1.1. These first ten genes are ordered according to their log2FoldChange values in order of the absolute value in the change of their expression levels. The results are ten different genes that displayed the most change in their gene expression levels regardless of being upregulated or being downregulated. The *log2FoldChange* value illustrates the change of expression the best by describing the magnitude of the change experienced in addition to the direction of the change experienced for each of these ten genes. There is a mix of upregulated (7 genes) and downregulated (3 genes) genes that are shown in the results figure. A quick explanation of each gene acronym from brief research can be seen below:

Short Gene Acronym Descriptions:

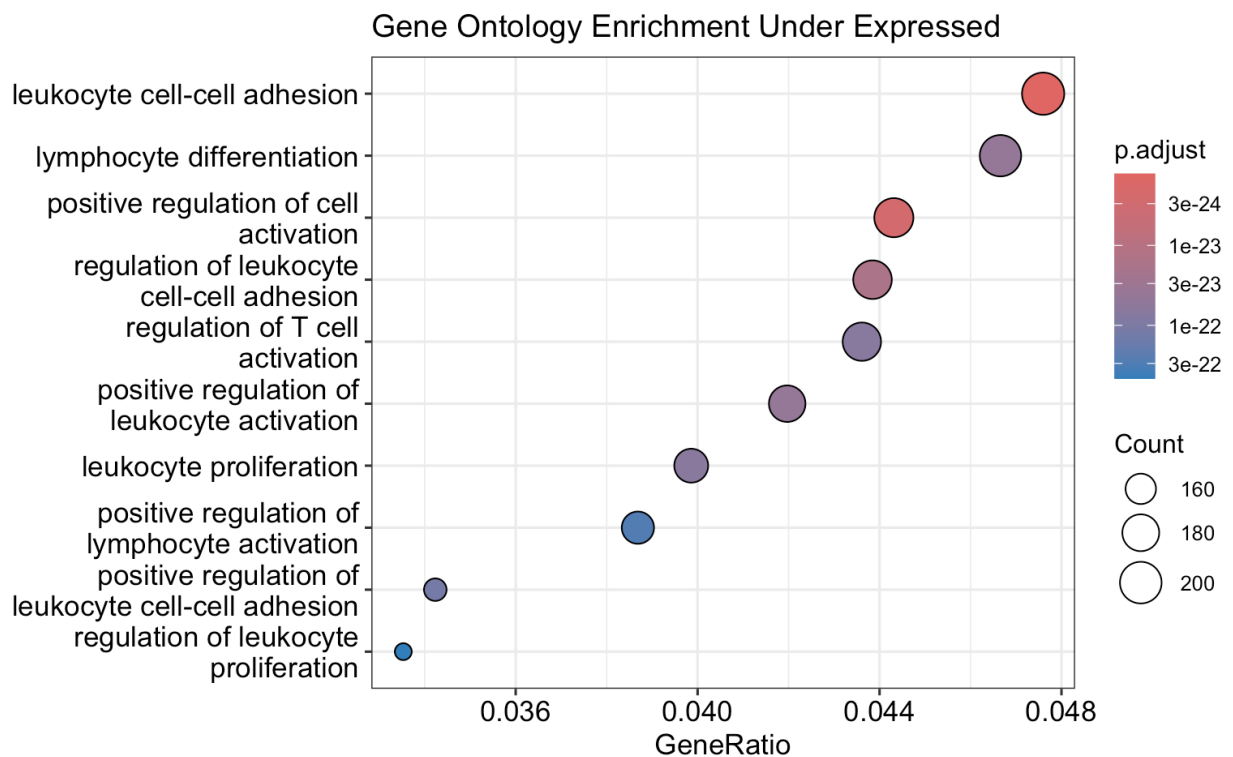
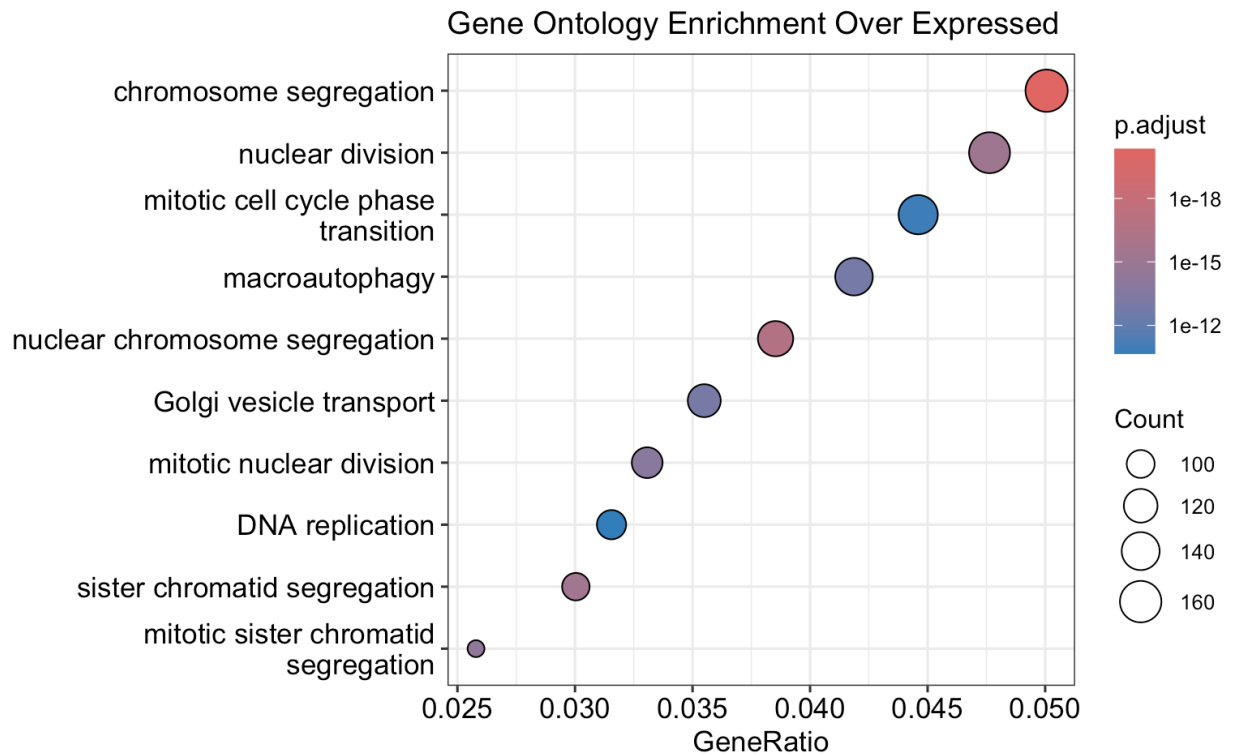
CSN2 -> part of the beta casein family; a primary milk protein secreted in human milk

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SPANXA2 -> part of the Sperm Protein Associated Nucleus X-linked family that are  
Cancer-Testis-Antigens (CTAs)  
GAGE12D -> part of the GAGE family; highly expressed in CTAs  
SPANXC -> member of SPANX family and also CTAs  
GAGE2B -> member of GAGE family; highly expressed in CTAs  
CSN3 -> part of beta casein family; another milk protein secreted  
FAM9C -> not a lot known aside from arising through duplication on X chromosome  
PNMT -> enzyme that helps with conversion of norepinephrine to epinephrine  
GAGE4 -> member of GAGE family; highly expressed in CTAs  
LALBA -> another milk protein important to protein lactose synthase
```

The importance of these upregulated and downregulated genes can be directly linked to their functionality that may ultimately be affecting the breast cancer of the patients in the dataset. At a surface level look at one of the genes, CSN2, the gene can be seen as a direct link to functionality of the breast area since it codes for a protein that is known to function in human milk. Understanding the different functionalities of the breast area that are being affected can help provide useful understanding in terms of the effects of the patient's breast cancer.

Pathway Enrichment – Header 1.3

```
Gene Ontology Key:  
X-axis -> Gene Ontology terms  
Y-axis -> gene ratio belonging to the specific term  
padj -> adjusted p value to consider many statistical tests applied  
Count -> corresponding value of genes mapping to the Gene Ontology term
```



The Gene Ontology figures display the results for the first pathway enrichment analysis which aims to find the representative biological processes and pathways that are being affected

by over-expressed genes or under-expressed genes from the differential expression analysis discussed in Headers 1.1 and 1.2. Separating the significant differentially expressed genes by their log2FoldChange level being greater than zero (over-expressed) or less than zero (under-expressed), and plotting according to the corresponding biological process in the Gene Ontology database was how the results were obtained for these figures. Both figures display the biological processes that are being affected by the over-expression or under-expression of the differentially expressed genes. Each of the biological processes are displayed on the plot through the point's color that embodies the p-adjusted value and its size that embodies the count of expression. The biological processes that are regulated by the largest number of the differentially expressed genes corresponds to the count variable, which for the over-expressed genes was chromosome segregation and for under-expressed genes was leukocyte cell-cell adhesion. The importance of these pathways accounting for largest proportion and absolute number of over/under expressed genes is that they can be identified as useful biological processes that represent the various functionalities of the differentially expressed genes derived from the patients with ERBB2 amplified breast cancer. Their functionality can be used to better explain processes seen in ERBB2 amplified cancer in general.

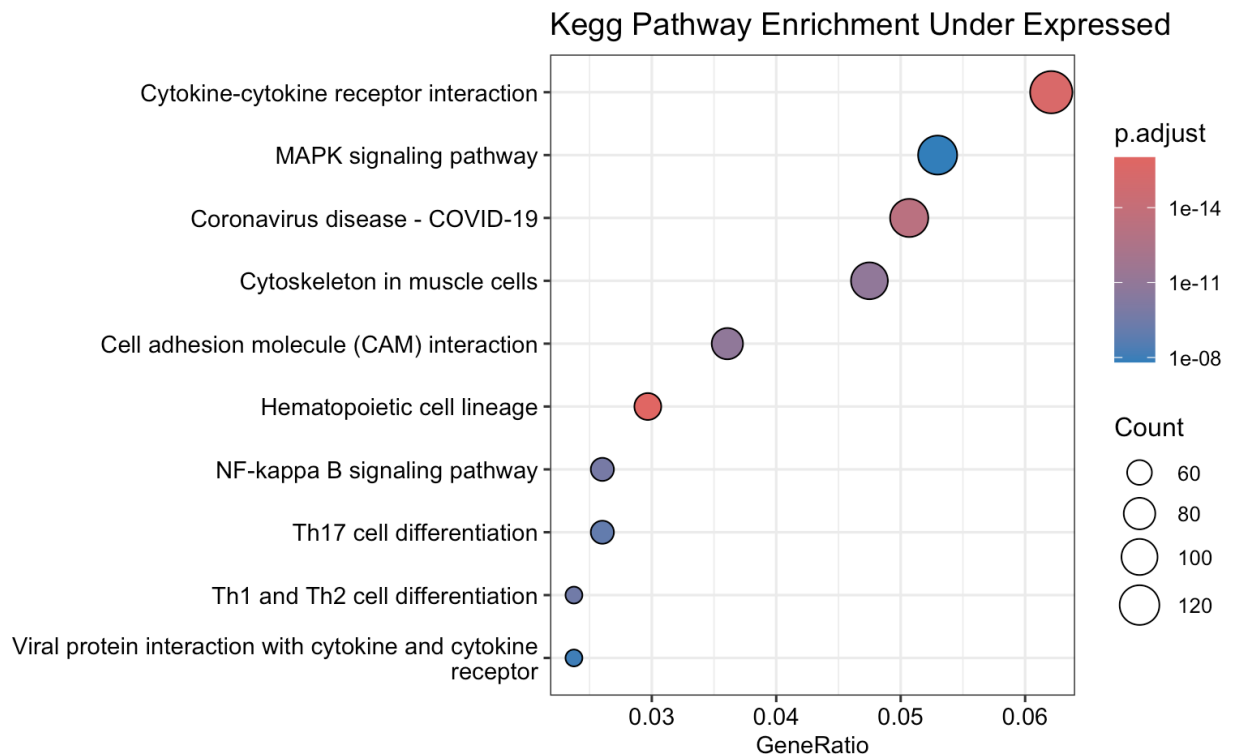
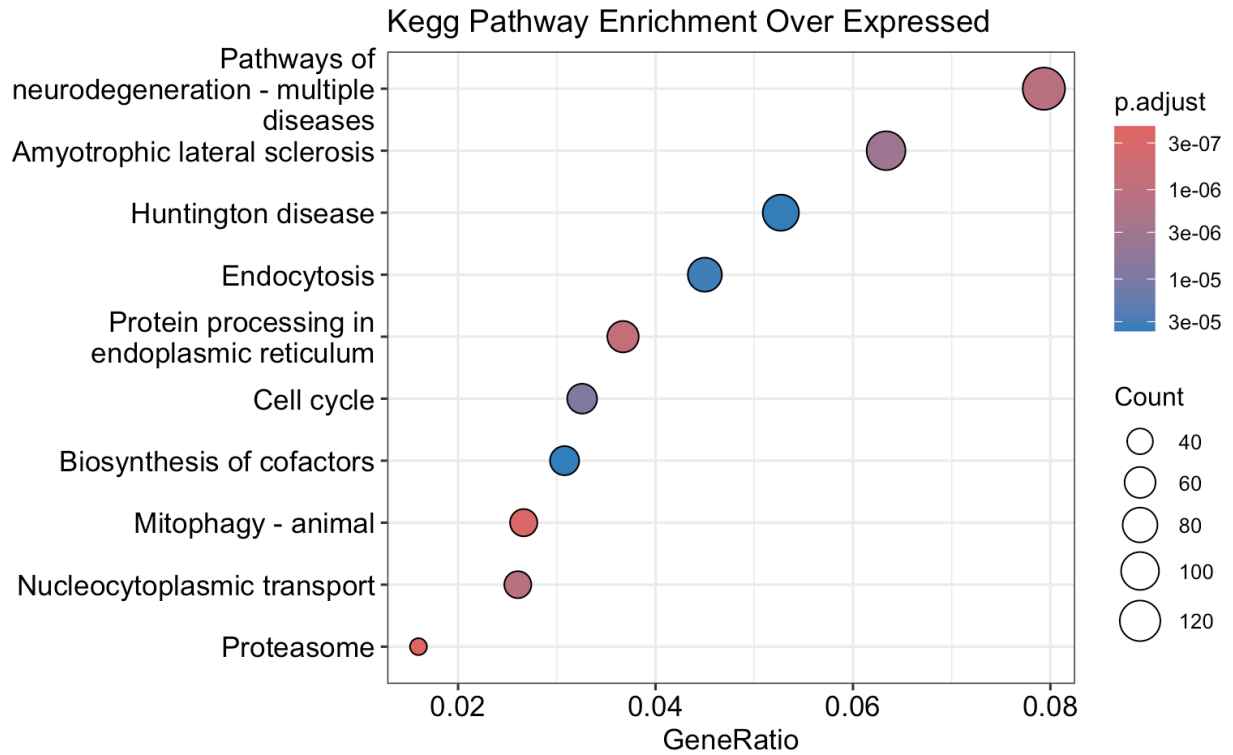
KEGG Pathway Key:

X-axis -> KEGG specific pathway terms

Y-axis -> gene ratio belonging to the specific term

padj -> adjusted p value to consider many statistical tests applied

Count -> corresponding value of genes mapping to the pathway term



The KEGG pathway analysis provides two figures representing the pathways that can be affected by the over-expressed or under-expressed genes of the differential expression analysis in general. The same over/under expressed genes that were separated as the significant

differentially expressed genes according to their log2FoldChange level were used for this analysis. The analysis provided a different basis - the specific pathways provided through KEGG – that were identified through their gene symbol (Hugo Symbol). The pathways shown to be regulated by the largest number of the differentially expressed genes that were identified for over-expressed genes were the general pathways of neurodegeneration – multiple diseases. For the under-expressed genes, the pathway was the Cytokine-cytokine receptor interaction. The importance of these pathways accounting for largest proportion and absolute number of over/under expressed genes is not specifically related to the cause of ERBB2 amplified breast cancer of the patients in this dataset, however, their functionality can be generalized and useful as these genes are present in the ERBB2 amplified breast cancer patients. This means the functionality of the genes could exert similar amplified proliferation functionality in the patients of this dataset, although, the pathways seen here are not being affected in the patients.

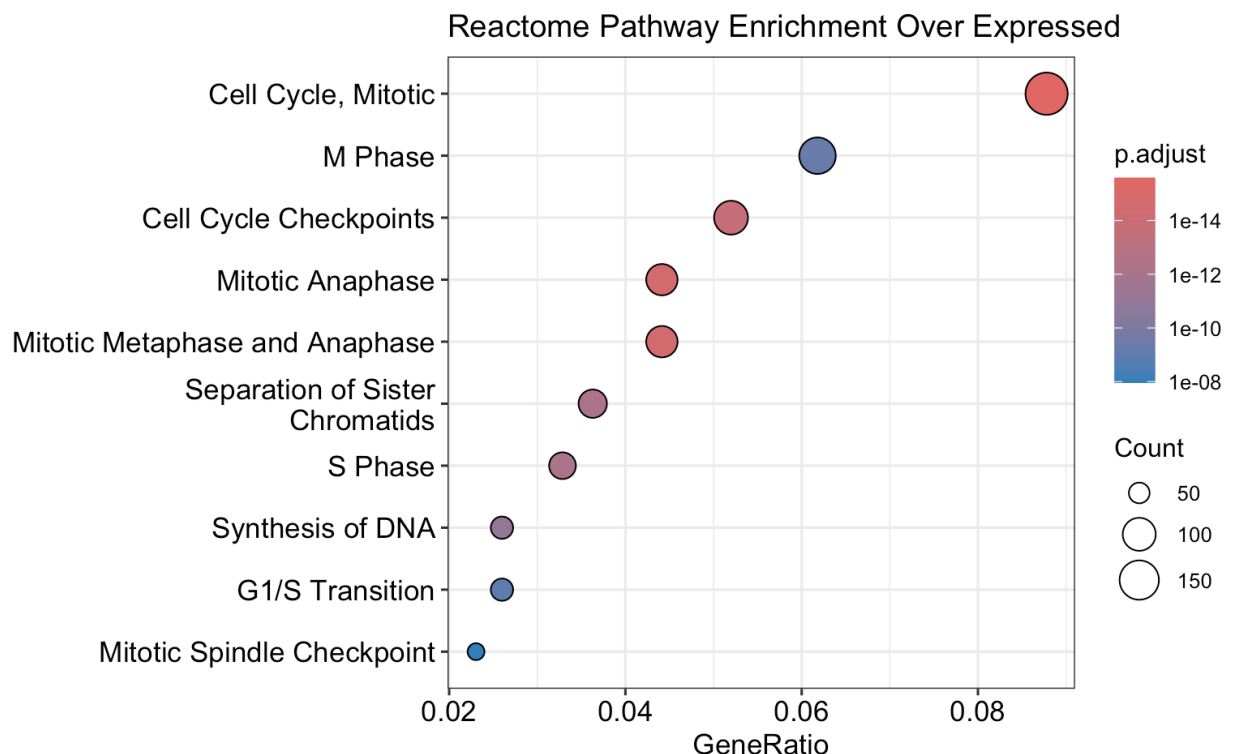
Reactome Pathway Key:

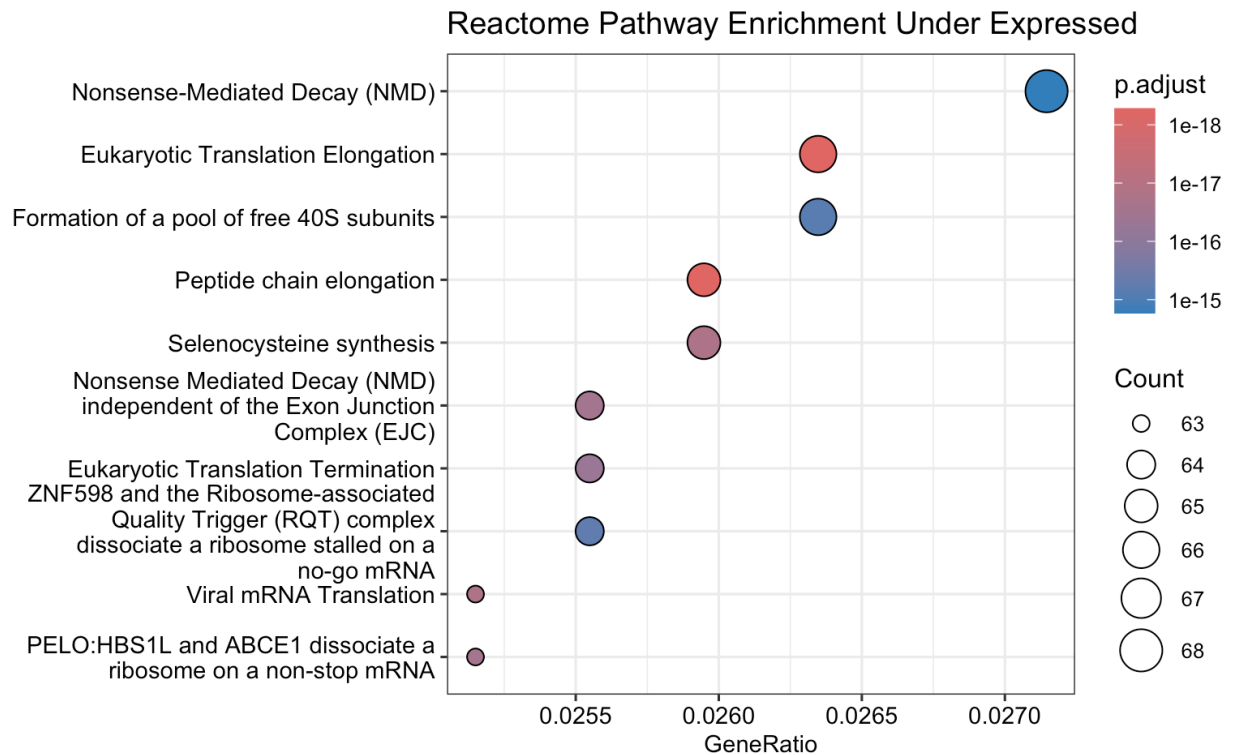
X-axis -> Reactome specific pathway terms

Y-axis -> gene ratio belonging to the specific term

padj -> adjusted p value to consider many statistical tests applied

Count -> corresponding value of genes mapping to the pathway term

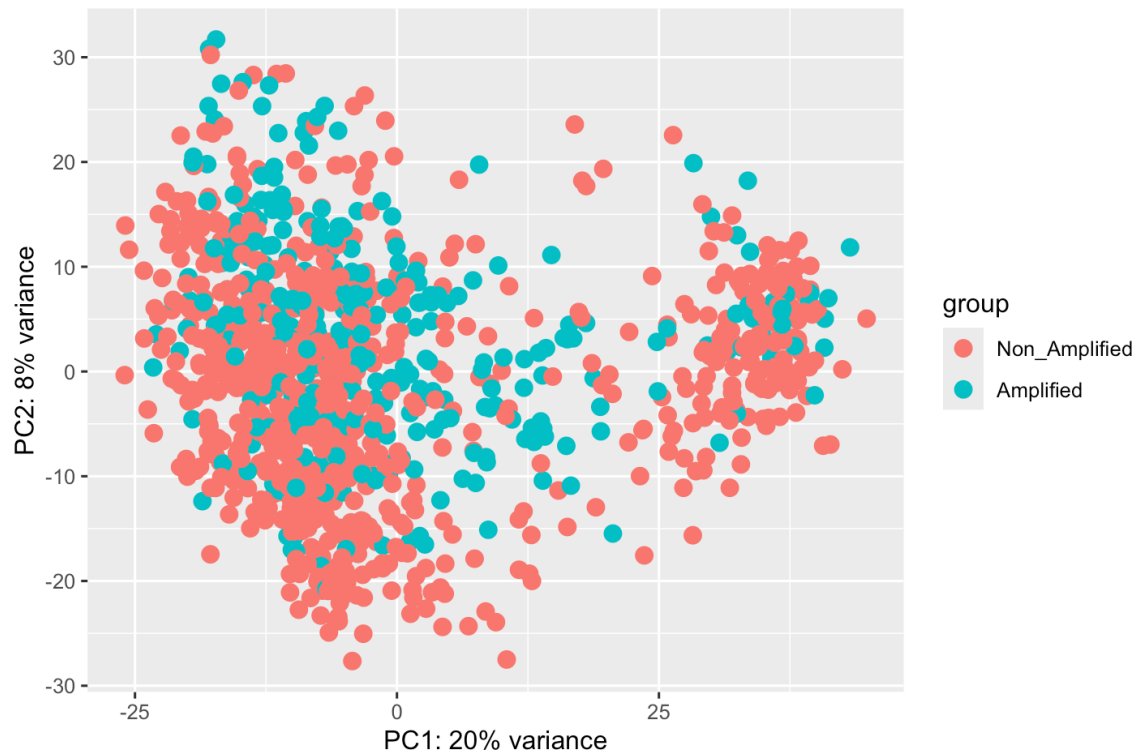




The Reactome pathway analysis provides two additional figures that represent the pathways that can be affected by the over-expressed or under-expressed genes from the differential expression analysis. The same over/under expressed genes separated as the significant differentially expressed genes according to their log2FoldChange level were used for this analysis. The analysis provided another basis of specific pathways through the Reactome database and were identified through their entrez id corresponding to each of the genes. The pathways shown to be regulated by the largest number of the differentially expressed genes that were identified for the over-expressed genes were the general phases of the cell cycle and more specifically the Mitotic phase. For the under-expressed genes, it was the Nonsense-Mediated Decay (NMD) pathway. The importance of these pathways accounting for the largest proportion and absolute number of over/under expressed genes is that for the cell cycle / Mitotic phase, it reflects previous observations that were made about pathway outcomes like proliferation that were discussed to have similar effects to on the pathways that are present in ERBB2 amplified breast cancer. As for the process of NMD, it is known as a defense system that helps control the quality of mRNA and prevents the production of harmful proteins. With this knowledge, the harmful production of proteins could be a significant issue that is also present in ERBB2 amplified breast cancer.

Principal Component Analysis (PCA) Plot – Header 1.4

Principal Component Analysis Key:
PC -> Principal Component
Group -> ERBB2 status

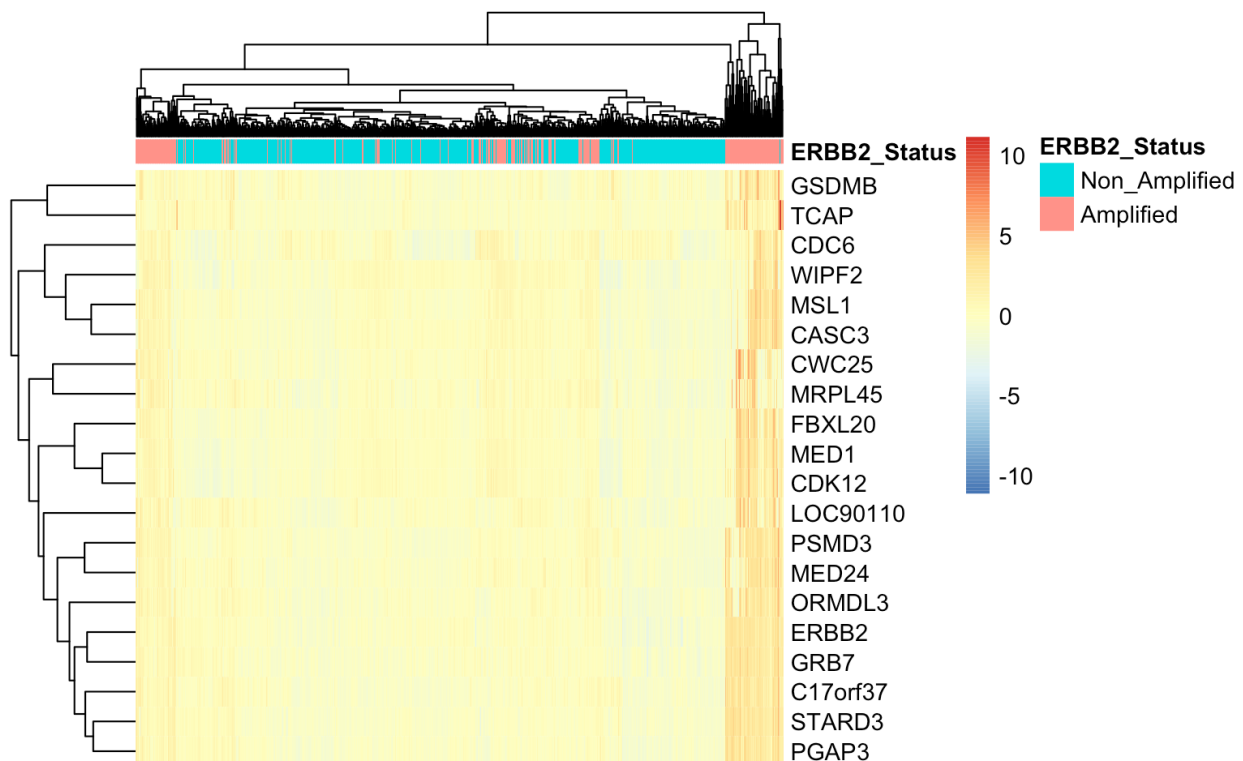


The principal component analysis was completed by first computing the variance stabilized transformed (vst) expression values from the differentially expressed genes which was explained under Header 1.1. These vst values are transformed across the wide range of raw RNA expression levels that were used to perform the differential expression analysis. Using these vst values, the principal component analysis was performed and grouped according to the ERBB2 amplified or non-amplified expression levels. The plot shows the two principal components that explain the most variance from these vst values, where PC1 explains 20% of variance and PC2 explains 8% of variance. The combined 28% of variance represents a minority share of variance and means that the missing 72% of variance makes up the majority of the explanation for the variation and more specifically the explanation of the differences in the breast cancer across the patients from the datasets. Visually, the PCA plot

reflects this statement as the two groups are not very separable, although, the separation that can be seen is due to PC1 which makes up a large share of variance with 20% of total variance. Overall, this is important because ERBB2 amplification is shown to be identified as a contributor to the overall diagnosis of breast cancer, however it is not attributed as the dominant factor or sole cause of breast cancer in these patients.

Heatmap – Header 1.5

Heatmap Key:
 ERBB2_Status -> ERBB2 status of the patients, Amplified or Non-amplified
 Gene acronyms -> Gene acronyms for the top 20 differentially expressed genes
 Color bar scale -> z-score statistic showing implying direction of expression levels



The heatmap plot was computed using the vst values of the differentially expressed genes explained in figure 1.3. Using these vst values, the separation of values was created based on the top twenty values that were the most statistically significant according to their adjusted p-value, as opposed to the log2FoldChange ordering that was applied for the top ten ranking of the differential expression genes under Header 1.2. Considering this, these genes do not represent the

largest change in expression levels, but do represent the most statistically significant genes with the largest statistical certainty in actual representation of real change - as opposed to being changes that are random occurrences. Visually, there are groups of genes that are noticeably appearing as over-expressed and amplified as seen with the cluster of amplified bars on the righthand side of the figure displayed with a darker color. The density of the color represents the z-score statistic that reflects the direction of expression as upregulated for all of these genes. The dendrograms represent the items being compared which are the patients in the top dendrogram and the genes in the left dendrogram. The branches of each represent the groups that share similar expression profiles and expression patterns for each dendrogram, respectively. The heatmap shows the distinction between ERBB2 amplified vs non-amplified patients along with the genes that are most certain to be present in ERBB2 amplification which is important in understanding the biological functional changes that have been shown throughout these analyses.

Lasso Regularized Cox Regression with DE Genes – Header 1.6

Gene Acronym	Cox's Regression Coefficient
APOA5	0.103502797
ARID1B	0.084265221
BTBD6	-0.094779859
C6orf138	0.081640436
C6orf81	-0.071960514
C7orf29	-0.059304358
C8orf33	0.038018721
CCNA1	-0.051059813
CD24	0.019151889
CEL	0.211111265
CLCA4	-0.129964294
CPLX4	-0.142654661
CYTL1	0.154494926
DDAH1	-0.030093082
DLGAP2	0.197826838
FAM179A	-0.160893353
FOXJ1	-0.003652026
GREB1L	-0.013769204
GRIA3	0.086352247
GSTM4	-0.010788249
HDGFL3	-0.012169924
IGJ	-0.031250021
IRF2	-0.130689234

KLHDC7B	-0.016500315
KLRB1	-0.113056296
KRT12	0.033517957
MAPT-AS1	-0.141305843
GALNT9-AS1	0.037915408
LOC143188	0.166445960
LRRC14B	0.014537418
LRRC8D	-0.073958327
<u>MCTS1</u>	<u>0.245593549</u>
NFE2	-0.107326575
NOS1	0.057818502
NTRK3	-0.017479636
NUP43	0.035620299
OR52E6	0.388803693
OR7A5	-0.049809884
OR7C1	-0.083324755
OR7D2	-0.034732875
PCDHGA11	0.123101594
PCMT1	0.059899461
PDP1	0.020235736
PELO	0.068865763
PGK1	0.083991121
QPRT	0.012974377
QRFPR	-0.094099544
RAPGEFL1	-0.088409109
RPH3A	0.059145999
RPL19	-0.010060860
SAV1	-0.121912439
SERPINA1	-0.038854963
SFTPB	-0.073324848
SHBG	-0.403201860
SLC16A2	0.009373491
SLC1A4	-0.048586700
SMR3B	0.022326538
ST7OT1	-0.111528440
TAPBPL	-0.049625115
TMEM14B	-0.033997522
TOX3	-0.003345098
TPRXL	0.031883472
TRMT5	-0.082526303
TTC19	0.004451519
TXNDC6	-0.227553822
WNT3A	0.084920007
XRCC4	0.177359696
ZNF498	-0.038712154

The table above shows the results from the overall survival model's prognostic gene signature for the breast cancer dataset. Using the vst values of the differentially expressed genes, the genes above were selected based on their ability to predict the overall survival risk for the patients. In combination with the overall survival status and overall survival in months, Cox's model using Lasso regression provided this list of genes and proved its ability as a capable non-parametric model for survival analysis. Along with each of the genes are the cox's regression coefficients which are viewed as the hazard ratio or risk of the event which in this case is death. The table shows a wide range of Cox's regression coefficients for the genes which illustrate the existence of risk factors like MCTS1 that has a coefficient of +0.246 indicating a high association with poor prognosis for breast cancer based on this dataset. Overall, the survival model provides another way to evaluate the impact that the differentially expressed genes have on these ERBB2 amplified breast cancer patients.

Discussion:

One of the key pathways that was identified in the KEGG pathway enrichment analysis was the MAPK pathway. This pathway was found through the KEGG pathway enrichment analysis of the under-expressed genes that came from the differential expression analysis of ERBB2. The MAPK pathway, formally known as the Mitogen-activated protein kinase pathway, is responsible for the amplification and transmission of signals that are involved in key cellular processes like cell proliferation and cell death.⁷ Activation of the MAPK pathway comes from different ligands that are specific for protein receptors like the growth factor, steroid hormone, and G protein receptors.⁸ As referenced in *The role of mitogen-activated-protein (MAP) kinase in breast cancer* by Santen et Al., MAPK-ERK-1 and MAPK-ERK-2 pathways are recognized as key sub-pathways that have been stated to be the most relevant to breast cancer - although the authors state that the MAPK pathway is typically amplified in breast cancer. This is contradictory to what was observed in the results of the KEGG pathway enrichment analysis.⁹ In

⁷ Santen, R.J. *et al.* (2002) "The role of mitogen-activated protein (MAP) kinase in breast cancer," *The Journal of Steroid Biochemistry and Molecular Biology*, 80(2), pp. 239–256. Available at: [https://doi.org/10.1016/S0960-0760\(01\)00189-3](https://doi.org/10.1016/S0960-0760(01)00189-3).

⁸ (Santen *et al.*, 2002)

⁹ (Santen *et al.*, 2002)

the analysis from the KEGG Pathway Enrichment Under Expressed plot under the Header 1.3, we can see that the MAPK pathway holds the second largest proportion in gene ratio (~0.055 gene ratio) and absolute number of under expressed genes. This is contradictory as the understanding that has been established and been the main discussion of analysis has focused on the amplification of ERBB2 that as stated in the Introduction, is supposed to be a driver of proliferation through pathways like the MAPK pathway in normal and mutated functionality. So, the expected functionality of the MAPK pathway when ERBB2 is amplified would be that the pathway is amplified too. This is because the amplified levels of ERBB2 are expected to lead to higher-than-normal ERBB2 levels that result in increased dimerization with HER3 and stronger signaling to pathways like MAPK. These stronger signals would lead to increased proliferation and therefore tumor growth. However, this functionality of the MAPK pathway is not seen although ERBB2 amplification is present.

Although the expectation that ERBB2 amplification would lead to a pathway like MAPK becoming amplified due the direct interaction between the two is untrue, there were clear signals that amplification of other processes and pathways that caused increased proliferation were present. These pathways were highlighted in the opposing pathway enrichment analysis, the KEGG Pathway Enrichment Overexpressed plot. The general pathways of neurodegeneration - multiple diseases that accounted for the largest proportion and absolute number of over expressed genes confirms that there exists an increase in the levels of genes that share a similar functionality which in this case is amplified proliferation. One of the reasons that could help explain this activity was referenced in *The role of mitogen-activated-protein (MAP) kinase in breast cancer*, and explained as the potential crosstalk between the MAPK and other pathways like the PI3K/AKT pathway for example. In general, cross talk was mentioned as possible contributing factor to the potential increase and decrease between tradeoffs of signaling pathways' signal intensity. This could be part of the solution as to why the MAPK pathway would be showing an under-expressed level of differentially expressed genes despite intuition expecting over-expressed levels.¹⁰ In the future analyses, this interaction between amplified ERBB2 and under-expressed levels of its differentially expressed genes in the MAPK pathway would be interesting to investigate further.

¹⁰ (Santen *et al.*, 2002)

Link to GitHub: <https://github.com/Ant-Small/Gene-Expression-and-Analysis-Interpretation>

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